

OS LAB-2

1. Write a C program to simulate SJF scheduling algorithm non pre-emptive

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>
#define n 5

int ct_1=0;
float tat_1=0,wt_1=0;

typedef struct{
    int bt;
    int at;
    int ct;
    int tat;
    int wt;
}process;

process a[n]={0};

int main(){

for(int i=0;i<n;i++){
    printf("enter at, bt for process %d \n",i+1);
    scanf("%d",&a[i].at);
    scanf("%d",&a[i].bt);
}

for(int i=0;i<n-1;i++){
    for(int j=i+1;j<n;j++){
        if(a[i].at > a[j].at){
            process temp=a[i];
            a[i]=a[j];
            a[j]=temp;
        }
    }
}

for(int i=0;i<n-1;i++){
    for(int j=i+1;j<n;j++){
        if(a[i].at==a[j].at && a[i].bt>a[j].bt){
```

```

        process temp=a[i];
        a[i]=a[j];
        a[j]=temp;
    }
}

for(int i=0;i<n;i++){

    if(ct_1 < a[i].at)
        ct_1 = a[i].at;

    ct_1 += a[i].bt;

    a[i].ct=ct_1;
    a[i].tat=a[i].ct-a[i].at;
    a[i].wt=a[i].tat-a[i].bt;

    tat_1+=a[i].tat;
    wt_1+=a[i].wt;
}

printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");

for(int i=0;i<n;i++){
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,a[i].at,a[i].bt,a[i].ct,a[i].tat,a[i].wt);
}

printf("average tat=%f,avg wt=%f",tat_1/n,wt_1/n);

}

```

OUTPUT:

```
enter at, bt for process 1
0 3
enter at, bt for process 2
2 2
enter at, bt for process 3
3 5
enter at, bt for process 4
4 4
enter at, bt for process 5
6 1

Process AT      BT      CT      TAT      WT
P1      0       3       3       3       0
P2      2       2       5       3       1
P3      3       5      10       7       2
P4      4       4      14      10       6
P5      6       1      15       9       8
average tat=6.400000,avg wt=3.400000
```

2. Write a C program to simulate SJF scheduling algorithm preemptive.

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define n 5

int ct_1 = 0;
float tat_1 = 0, wt_1 = 0;

typedef struct{
    int bt;
    int at;
    int ct;
    int tat;
    int wt;
    int rt;
}process;

process a[n] = {0};

int main(){

    for(int i=0;i<n;i++){
        printf("Enter AT and BT for process %d\n",i+1);
        scanf("%d",&a[i].at);
        scanf("%d",&a[i].bt);
        a[i].rt = a[i].bt;
    }

    int completed = 0;
    int time = 0;
    int min_index = -1;
    int min_bt = 9999;

    while(completed < n){

        min_bt = 9999;
        min_index = -1;

        for(int i=0;i<n;i++){
            if(a[i].at <= time && a[i].rt > 0){
```

```

        if(a[i].rt < min_bt){
            min_bt = a[i].rt;
            min_index = i;
        }
    }
}

if(min_index == -1){
    time++;
    continue;
}

a[min_index].rt--;
time++;

if(a[min_index].rt == 0){
    completed++;

    a[min_index].ct = time;
    a[min_index].tat = a[min_index].ct - a[min_index].at;
    a[min_index].wt = a[min_index].tat - a[min_index].bt;

    tat_1 += (float)a[min_index].tat / n;
    wt_1 += (float)a[min_index].wt / n;
}
}

printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");

for(int i=0;i<n;i++){
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,a[i].at,a[i].bt,a[i].ct,a[i].tat,a[i].wt);
}

printf("\nAverage TAT = %f",tat_1);
printf("\nAverage WT = %f\n",wt_1);
}

```

OUTPUT:

Enter AT and BT for process 1

0 3

Enter AT and BT for process 2

2 2

Enter AT and BT for process 3

3 5

Enter AT and BT for process 4

4 4

Enter AT and BT for process 5

6 1

Process	AT	BT	CT	TAT	WT
P1	0	3	3	3	0
P2	2	2	5	3	1
P3	3	5	15	12	7
P4	4	4	10	6	2
P5	6	1	7	1	0

Average TAT = 5.000000

Average WT = 2.000000

3. Write a C program to simulate priority scheduling algorithm non preemptive.

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define n 5

int ct_1 = 0;
float tat_1 = 0, wt_1 = 0;

typedef struct{
    int bt;
    int at;
    int pr;
    int ct;
    int tat;
    int wt;
    bool done;
}process;

process a[n] = {0};

int main(){

    for(int i=0;i<n;i++){
        printf("Enter AT, BT and Priority for process %d\n",i+1);
        scanf("%d",&a[i].at);
        scanf("%d",&a[i].bt);
        scanf("%d",&a[i].pr);
        a[i].done = false;
    }

    int completed = 0;
    int time = 0;

    while(completed < n){

        int min_pr = 9999;
        int index = -1;

        for(int i=0;i<n;i++){
            if(a[i].at <= time && a[i].done == false){
                if(a[i].pr < min_pr){
```

```

        min_pr = a[i].pr;
        index = i;
    }
}

if(index == -1){
    time++;
    continue;
}

time += a[index].bt;
a[index].ct = time;

a[index].tat = a[index].ct - a[index].at;
a[index].wt = a[index].tat - a[index].bt;

tat_1 += (float)a[index].tat / n;
wt_1 += (float)a[index].wt / n;

a[index].done = true;
completed++;
}

printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");

for(int i=0;i<n;i++){
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,a[i].at,a[i].bt,a[i].pr,a[i].ct,a[i].tat,a[i].wt);
}

printf("\nAverage TAT = %f",tat_1);
printf("\nAverage WT = %f",wt_1);

}

```

OUTPUT:

Enter AT, BT and Priority for process 1

0 3 1

Enter AT, BT and Priority for process 2

2 2 2

Enter AT, BT and Priority for process 3

3 5 3

Enter AT, BT and Priority for process 4

4 4 4

Enter AT, BT and Priority for process 5

6 1 5

Process	AT	BT	PR	CT	TAT	WT
P1	0	3	1	3	3	0
P2	2	2	2	5	3	1
P3	3	5	3	10	7	2
P4	4	4	4	14	10	6
P5	6	1	5	15	9	8

Average TAT = 6.400001

Average WT = 3.400000

4. Write a C program to simulate priority scheduling algorithm preemptive.

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define n 5

int ct_1 = 0;
float tat_1 = 0, wt_1 = 0;

typedef struct{
    int bt;
    int at;
    int pr;
    int ct;
    int tat;
    int wt;
    int rt;
}process;

process a[n] = {0};

int main(){

    for(int i=0;i<n;i++){
        printf("Enter AT, BT and Priority for process %d\n",i+1);
        scanf("%d",&a[i].at);
        scanf("%d",&a[i].bt);
        scanf("%d",&a[i].pr);
        a[i].rt = a[i].bt;
    }

    int completed = 0;
    int time = 0;

    while(completed < n){

        int min_pr = 9999;
        int index = -1;

        for(int i=0;i<n;i++){
            if(a[i].at <= time && a[i].rt > 0){
                if(a[i].pr < min_pr){
```

```

        min_pr = a[i].pr;
        index = i;
    }
}

if(index == -1){
    time++;
    continue;
}

a[index].rt--;
time++;

if(a[index].rt == 0){
    completed++;

    a[index].ct = time;
    a[index].tat = a[index].ct - a[index].at;
    a[index].wt = a[index].tat - a[index].bt;

    tat_1 += (float)a[index].tat / n;
    wt_1 += (float)a[index].wt / n;
}

printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");

for(int i=0;i<n;i++){
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,a[i].at,a[i].bt,a[i].pr,a[i].ct,a[i].tat,a[i].wt);
}

printf("\nAverage TAT = %f",tat_1);
printf("\nAverage WT = %f",wt_1);

}

```

OUTPUT:

Enter AT, BT and Priority for process 1

0 3 1

Enter AT, BT and Priority for process 2

2 2 2

Enter AT, BT and Priority for process 3

3 5 3

Enter AT, BT and Priority for process 4

4 4 4

Enter AT, BT and Priority for process 5

6 1 5

Process	AT	BT	PR	CT	TAT	WT
P1	0	3	1	3	3	0
P2	2	2	2	5	3	1
P3	3	5	3	10	7	2
P4	4	4	4	14	10	6
P5	6	1	5	15	9	8

Average TAT = 6.400001

Average WT = 3.400000